

DBS Semester One Coursework

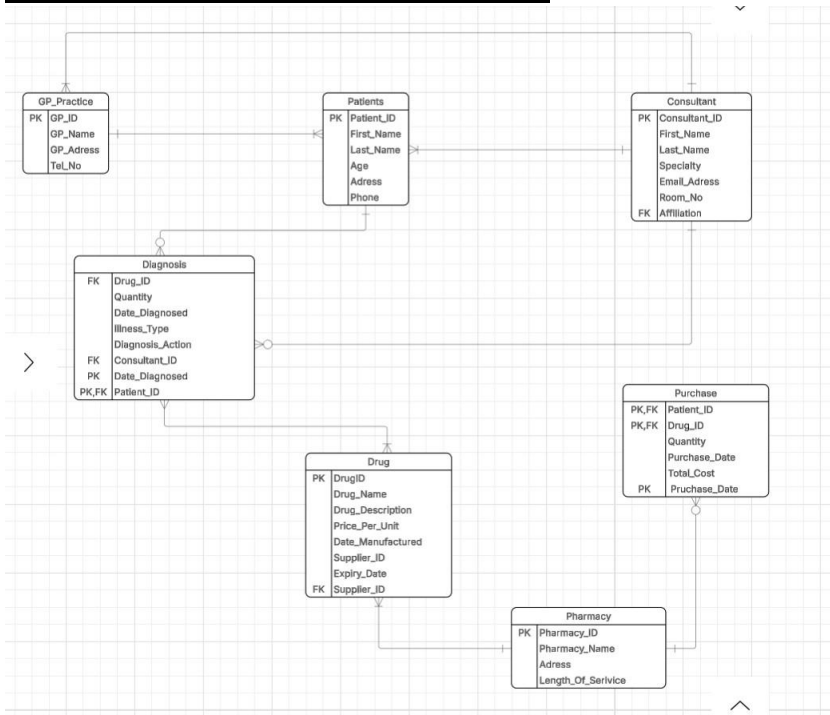
Database Systems

Module CRN: [N/A]

Roll No : [N/A]

User ID : [N/A]

E-R DIAGRAM (CROW'S FOOT NOTATION)



CREATE TABLE STATEMENTS:

1. CREATE TABLE Patients (

Patient_ID VARCHAR(10) PRIMARY KEY,
First_Name VARCHAR(50) NOT NULL,
Last_Name VARCHAR(50) NOT NULL,
Age Int NOT NULL,
Address VARCHAR(255),
Phone VARCHAR(20)
);

2. CREATE TABLE GP_Practice(

GP_ID VARCHAR(10) PRIMARY KEY,
GP_Name VARCHAR(50) NOT NULL,
Address VARCHAR(255) NOT NULL,
Tel_No VARCHAR(20) NOT NULL
);

3. CREATE TABLE Consultant (Consultant_ID

VARCHAR(10) PRIMARY KEY,
First_Name VARCHAR(50),
Last_Name VARCHAR(50),

Speciality VARCHAR(50) NOT NULL,
 Email_Address VARCHAR(255) NOT NULL
 UNIQUE,
 Room_No VARCHAR(10),
 Affiliation VARCHAR(10) NOT NULL,
 FOREIGN KEY (Affiliation) REFERENCES
 GP_Practice(GP_ID)
);

4. CREATE TABLE Drug (

Drug_ID VARCHAR(10) PRIMARY KEY,
 Drug_Name VARCHAR(50) NOT NULL,
 Drug_Description VARCHAR(50),
 Price_Per_Unit DECIMAL(10,2) NOT NULL,
 Date_Manufactured DATE NOT NULL,
 Supplier_ID VARCHAR(10) NOT NULL,
 Expiry_Date DATE CHECK(Expiry_Date >
 Date_Manufactured),
 FOREIGN KEY (Supplier_ID) REFERENCES
 Pharmacy(Pharmacy_ID)
);

5. CREATE TABLE Pharmacy (

Pharmacy_ID VARCHAR(20) PRIMARY KEY,
 Pharmacy_Name VARCHAR(100) NOT NULL,
 Address VARCHAR(255) NOT NULL,
 Length_Of_Service INT NOT NULL
);

6. CREATE TABLE Diagnosis (

INSERT INTO TABLE STATEMENTS:

Drug_ID VARCHAR(10) NOT NULL,
 Quantity INT NOT NULL,
 Date_Diagnosed DATE NOT NULL,
 Illness_Type VARCHAR(100) NOT NULL,
 Patient_ID VARCHAR(10) NOT NULL,
 Diagnosis_Action VARCHAR(20) NOT
 NULL,
 Consultant_ID VARCHAR(10) NOT NULL,
 PRIMARY KEY (Patient_ID,
 Date_Diagnosed),
 FOREIGN KEY (Drug_ID) REFERENCES
 Drug(Drug_ID),
 FOREIGN KEY (Patient_ID) REFERENCES
 Patient(Patient_ID),
 FOREIGN KEY (Consultant_ID) REFERENCES
 Consultant(Consultant_ID)
);

7. CREATE TABLE Purchase (

Patient_ID VARCHAR(10) NOT NULL,
 Drug_ID VARCHAR(10) NOT NULL,
 Quantity INT CHECK (0<Quantity),
 Purchase_Date DATE NOT NULL,
 Total_Cost Decimal(10,2),
 PRIMARY KEY (Patient_ID, Drug_ID,
 Purchase_Date),
 FOREIGN KEY (Patient_ID) REFERENCES
 Patients(Patient_ID),
 FOREIGN KEY (Drug_ID) REFERENCES
 Drug(Drug_ID)
);

1. INSERT INTO Patients (Patient_ID,First_Name, Last_Name, Age, Address, Phone)

VALUES

("PA1", "AMSA", "Zili", "35", "201A Apartment Road, Cape Town", "27-21-12128876765"),
("PA2", "Zoe", "Coye", "55", "33 Welbeck Ave, Cape Town", "27-21-10012424005"),
("PA3", "Joe", "Smiths", "30", "101 Trent Road, Cape Town", "27-21-90871829345"),
("PA4", "Aqua", "Moyo", "60", "Apartment 1A, Sandy Lane, Cape Town", "27-21-55022221581"),
("PA5", "Asali", "Zulu", "22", "153 FoxDale Crescent, Cape Town", "27-21-07923712880");

2. INSERT INTO GP_Practice (GP_ID, GP_Name, Address, Tel_No)

VALUES

("GP01", "Camda Health centre", "101 Buick Lane, Cape Town", "27-21-483392231"),
("GP02", "Lulabi Health centre", "12 Mill Drive, Cape Town", "27-21-483391931");

3. INSERT INTO Consultant (Consultant_ID, First_Name, Last_Name, Speciality, Email_Address, Room_No, Affiliation)

VALUES

("C1", "Alex", "Dana", "Cardiology", "a.dana1@email.com", "12A", "GP01"),
("C2", "Jade", "Kumo", "Psychiatry", "j.kumo1@email.com", "1A", "GP01"),
("C3", "Edith", "Jara", "Physiotherapy", "e.jara@email.com", "7", "GP02"),
("C4", "Mola", "Dude", "Ophthalmology", "m.due@email.com", "13", "GP02"),
("C5", "Agee", "Pamka", "Surgery", "a.pamka@email.com", "1", "GP02");

4. INSERT INTO Pharmacy (Pharmacy_ID, Pharmacy_Name, Address, Length_Of_Service)

VALUES

("Ph1", "ABC Pharma", "20 Acade Street, Cape Town", "20"),
("Ph2", "MediSupply Ltd", "16 Donoway Park, Cape Town", "5"),
("Ph3", "Camden Pharmacy", "5 Thianda Street, Cape Town", "15"),
("Ph4", "PJ Pharmacy", "15th Avenue, Downtown, Cape Town", "30"),
("Ph5", "Wanda Pharmacy", "100 Old Road, Cape Town", "25");

5. INSERT INTO Drug (Drug_ID, Drug_Name, Drug_Description, Price_Per_Unit, Date_Manufactured, Supplier_ID, Expiry_Date)

VALUES

("DR01", "Paracetamol", "local", 1.00, "2024-12-12", "Ph4", "2025-12-12"),
("DR02", "amoxycilin", "imported", 1.75, "2024-11-12", "Ph3", "2025-09-02"),

```
("DR03", "olanzapines", "imported", 1.75, "2024-11-01", "Ph2", "2025-05-12"),
("DR04", "Chloroquine", "imported", 1.75, "2023-11-01", "Ph1", "2026-05-12"),
("DR05", "ibuprofen", "local", 1.85, "2024-08-01", "Ph4", "2026-02-10");
```

6. INSERT INTO Diagnosis (Drug_ID, Quantity, Date_Diagnosed, Illness_Type, Patient_ID, Diagnosis_Action, Consultant_ID)

VALUES

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("DR01", 2, "2024-05-04", "Malaria", "PA1", "Referral", "C3"),
("DR02", 3, "2024-06-10", "ear infection", "PA2", "Referral", "C3"),
("DR03", 4, "2024-10-10", "Lung infection", "PA3", "Review", "C1"),
("DR04", 5, "2024-07-12", "psychotic review", "PA5", "Review", "C2");
```

7. INSERT INTO Purchase (Patient_ID, Drug_ID, Quantity, Purchase_Date, Total_Cost)

VALUES

```
("PA1", "DR01", "2", "2024-05-05", "2.00"),
("PA2", "DR05", "3", "2024-10-06", "5.55"),
("PA3", "DR05", "4", "2024-10-12", "7.40"),
("PA5", "DR03", "5", "2024-07-25", "8.75");
```

TABLE QUERY STATEMENTS:

1. SELECT *

FROM Drug

WHERE Drug_Name LIKE '%xy%'

OR Drug_Name LIKE '%za%';

2. SELECT Consultant_ID,

Patient_ID,

DATEDIFF(CURRENT_DATE, Date_Diagnosed) AS Days_Ago,

Illness_Type,

Date_Diagnosed,

Diagnosis_Action

FROM Diagnosis;

Comment : Most referrals seem to be handled by consultant C3.

3. SELECT patients.First_Name, patients.Last_Name,
SUM(Quantity) AS Total_Drugs_Bought
FROM Patients
JOIN Purchase ON Patients.Patient_ID = Purchase.Patient_ID
GROUP BY Patients.Patient_ID
HAVING SUM(Quantity) < 5
ORDER BY Total_Drugs_Bought DESC;

4. SELECT
Patient_ID,
SUM(Total_Cost) AS Total_Cost
FROM Purchase
GROUP BY Patient_ID
HAVING SUM(Total_Cost) < (
SELECT AVG(Total_Cost)
FROM (
SELECT SUM(Total_Cost) AS Total_Cost
FROM Purchase
GROUP BY Patient_ID)
AS Purchase_Totals
);